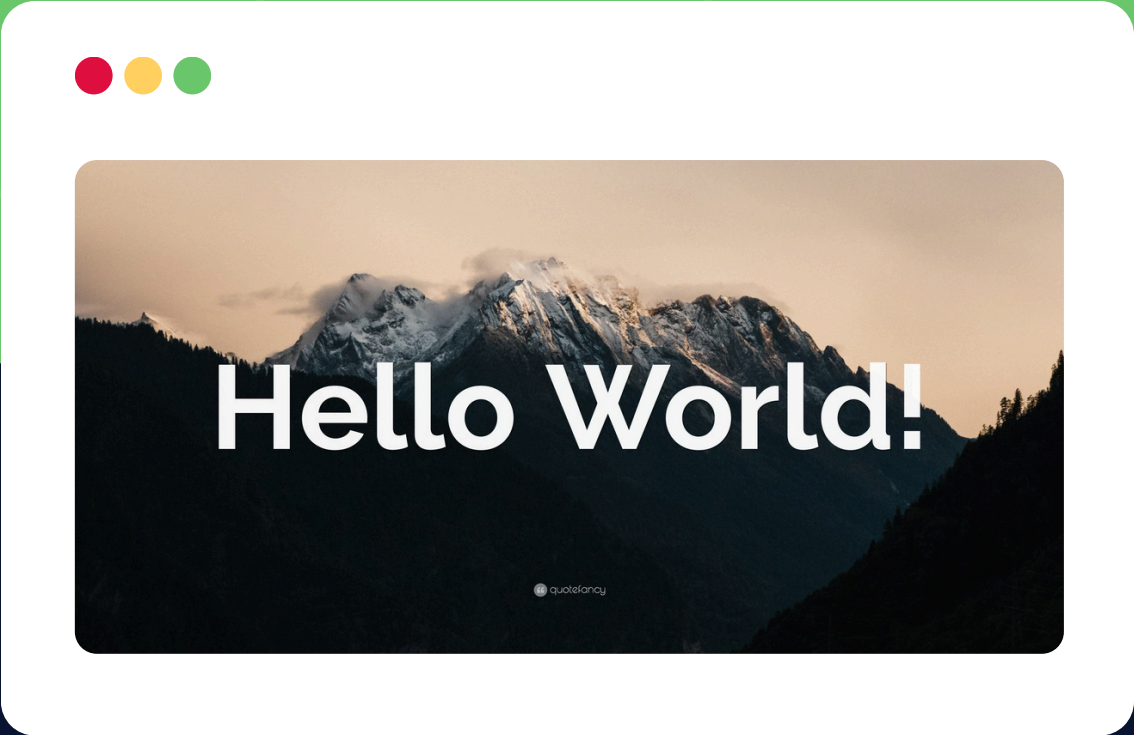
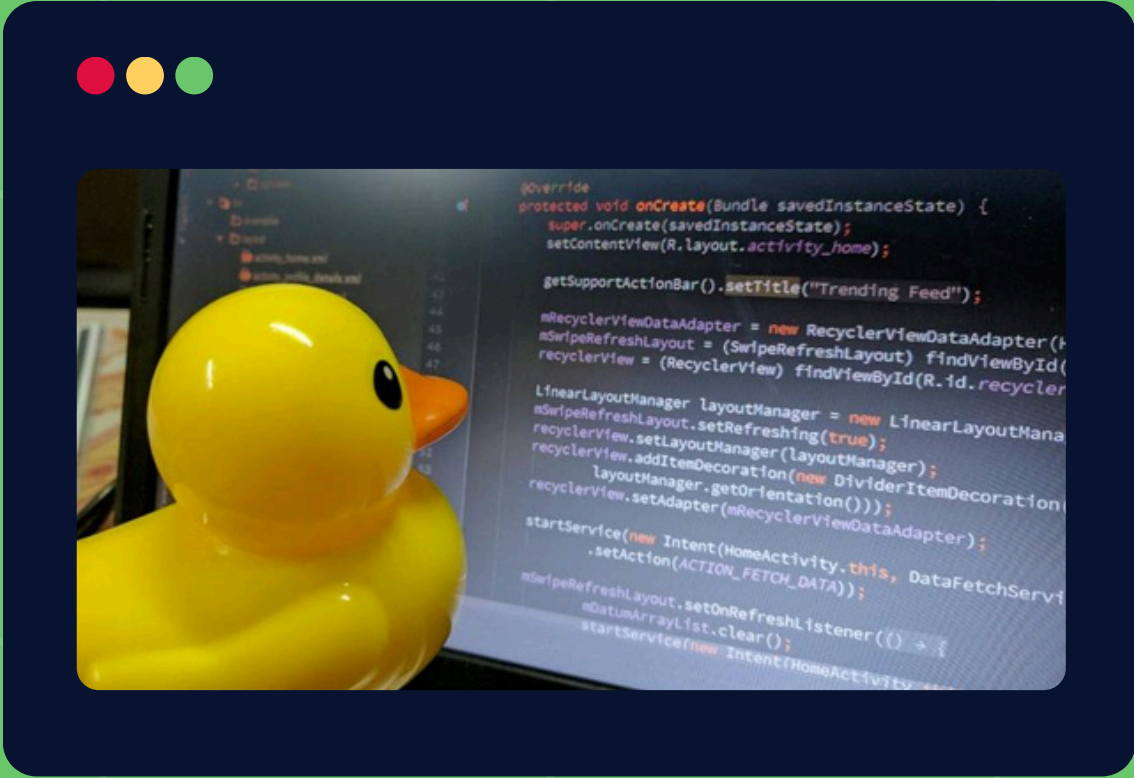




Hello World!



Decision Trees



Session -> 10



What are Decision Trees?

Decision Trees

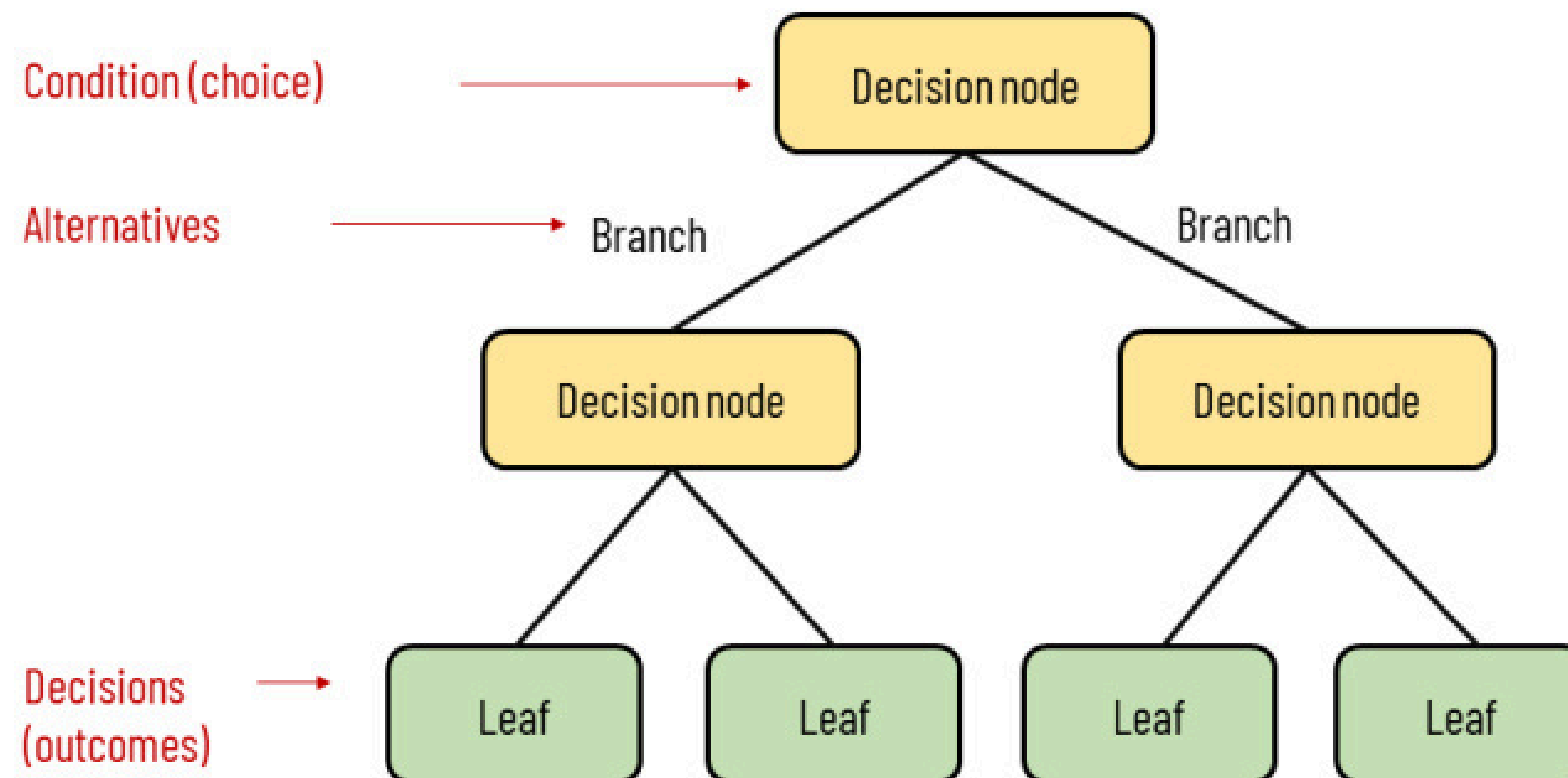
- Decision trees are a powerful supervised machine learning algorithm used for both **classification** and **regression** tasks.
- It models decisions based on a series of questions or conditions and their possible outcomes in a tree-like structure.
- Decision trees break down complex problems into a series of simple, interpretable steps.

Decision Tree Example



Decision Trees Components

Elements of a decision tree



Decision Trees Components

- **Root Node:** the topmost node of the tree. It contains the first question.
- **Decision Node:** Each question in our model is represented by a decision node.
- **Leaf Node:** A node that has no branches. It represents the decision we make after traversing the trees.
- **Branch:** A path that connects nodes in the tree. It represents the outcome of a decision or test at a decision node.
- **Depth:** The level or layer of a node within the tree, starting from the root node.

Entropy

What is entropy?

- Entropy measures the amount of uncertainty or disorder in the data.
- The more mixed the classes are, the higher the entropy.
- If all data points in a group belong to one class, entropy is zero (pure).

Lower entropy = more certainty, more information gained.

Entropy

Entropy Formula:

$$Entropy(S) = - \sum_{i=1}^c p_i \log_2(p_i)$$

- **S:** The feature/ dataset
- **c:** Number of classes
- **p_i:** The proportion of class i in the dataset

Entropy

Example:

If a dataset has:

- 5 positive samples (pass)
- 5 negative samples (fail)

$$Entropy = - \left(\frac{5}{10} \log_2 \frac{5}{10} + \frac{5}{10} \log_2 \frac{5}{10} \right) = 1$$

1 means **Maximum uncertainty**

(We cannot decide whether a student will pass or not based on that feature)



How Classification Trees Work?

Decision Tree Example

We want to classify whether the day is suitable for playing tennis or not. how to construct a decision tree for this problem?

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Decision Tree Example

First Step: choose the root node

- We choose the feature that gives us the **most certainty** (or the **least impurity**) about the target when we split the data by it.

How to choose it?

We calculate one of the following:

- **Information Gain:** Measures how much uncertainty is reduced after the split.
- **Gini Impurity:** Measures how “mixed” the classes are – lower is better.

Decision Tree Example

First Step: choose the root node

- **What is information gain?**

Information Gain is the difference between the uncertainty (entropy) before and after a split. In other words, it's like saying, "Did this split help us make better predictions?" **The higher the Information Gain, the better the split.**

$$IG(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} \times Entropy(S_v)$$

Decision Tree Example

First Step: choose the root node

- We use **entropy** to calculate the information gain:
1- Calculate the Entropy of the Whole Dataset
(This tells us how "uncertain" or "mixed" the data is before splitting).

$$\text{Entropy}(\text{Dataset}) = - \left(\frac{9}{14} \cdot \log_2 \frac{9}{14} \right) - \left(\frac{5}{14} \cdot \log_2 \frac{5}{14} \right)$$

Decision Tree Example

First Step: choose the root node

2- For Each Feature – Calculate the entropy for each variable

Feature(1): Outlook

Variables: sunny, overcast, rainy

1- Entropy for sunny (5 samples: 2 Yes, 3 No).

$$Entropy(Sunny) = - \left(\frac{2}{5} \log_2 \frac{2}{5} + \frac{3}{5} \log_2 \frac{3}{5} \right) = 0.971$$

Decision Tree Example

First Step: choose the root node

2- Entropy for Overcast (4 samples: 4 Yes, 0 No)

$$Entropy(Overcast) = - (1 \cdot \log_2 1 + 0 \cdot \log_2 0) = 0$$

3- Entropy for Rainy (5 samples: 3 Yes, 2 No)

$$Entropy(Rainy) = - \left(\frac{3}{5} \log_2 \frac{3}{5} + \frac{2}{5} \log_2 \frac{2}{5} \right) = 0.971$$

Decision Tree Example

First Step: choose the root node

Weighted Entropy for Outlook:

$$Entropy(Outlook) = \frac{5}{14} \cdot 0.971 + \frac{4}{14} \cdot 0 + \frac{5}{14} \cdot 0.971 = 0.694$$

Decision Tree Example

First Step: choose the root node

Feature(2): Temperature

Variables: Hot, Mild, Cool

1- Entropy for Hot (4 samples: 2 Yes, 2 No).

$$Entropy(Hot) = - \left(\frac{2}{4} \log_2 \frac{2}{4} + \frac{2}{4} \log_2 \frac{2}{4} \right) = 1.0$$

2- Entropy for Mild (6 samples: 4 Yes, 2 No).

$$Entropy(Mild) = - \left(\frac{4}{6} \log_2 \frac{4}{6} + \frac{2}{6} \log_2 \frac{2}{6} \right) = 0.918$$

Decision Tree Example

First Step: choose the root node

3- Entropy for Cool (4 samples: 3 Yes, 1 No).

$$Entropy(Cool) = - \left(\frac{3}{4} \log_2 \frac{3}{4} + \frac{1}{4} \log_2 \frac{1}{4} \right) = 0.811$$

Weighted Entropy for Temperature:

$$Entropy(Temp) = \frac{4}{14} \cdot 1 + \frac{6}{14} \cdot 0.918 + \frac{4}{14} \cdot 0.811 = 0.911$$

Decision Tree Example

First Step: choose the root node

Feature(3): Humidity

Variables: High, Normal

1- Entropy for High (7 samples: 3 Yes, 4 No).

$$Entropy(High) = - \left(\frac{3}{7} \log_2 \frac{3}{7} + \frac{4}{7} \log_2 \frac{4}{7} \right) = 0.985$$

2- Entropy for Normal (7 samples: 6 Yes, 1 No).

$$Entropy(Normal) = - \left(\frac{6}{7} \log_2 \frac{6}{7} + \frac{1}{7} \log_2 \frac{1}{7} \right) = 0.592$$

Decision Tree Example

First Step: choose the root node

Weighted Entropy for Humidity:

$$\textit{Entropy}(\textit{Humidity}) = \frac{7}{14} \cdot 0.985 + \frac{7}{14} \cdot 0.592 = 0.788$$

Decision Tree Example

First Step: choose the root node

Feature(4): Windy

Variables: True, False

1- Entropy for False (8 samples: 6 Yes, 2 No)

$$Entropy(False) = - \left(\frac{6}{8} \log_2 \frac{6}{8} + \frac{2}{8} \log_2 \frac{2}{8} \right) = 0.811$$

2- Entropy for True (6 samples: 3 Yes, 3 No)

$$Entropy(True) = - \left(\frac{3}{6} \log_2 \frac{3}{6} + \frac{3}{6} \log_2 \frac{3}{6} \right) = 1.0$$

Decision Tree Example

First Step: choose the root node

Weighted Entropy for Windy:

$$Entropy(Windy) = \frac{8}{14} \cdot 0.811 + \frac{6}{14} \cdot 1 = 0.892$$

Decision Tree Example

First Step: choose the root node

Choosing the feature with the highest information gain

1- Outlook:

$$\text{Gain(Outlook)} = 0.94 - 0.694 = \mathbf{0.246}$$



2- Temperature:

$$\text{Gain(Temperature)} = 0.94 - 0.911 = \mathbf{0.029}$$

3- Humidity:

$$\text{Gain(Humidity)} = 0.94 - 0.788 = \mathbf{0.152}$$

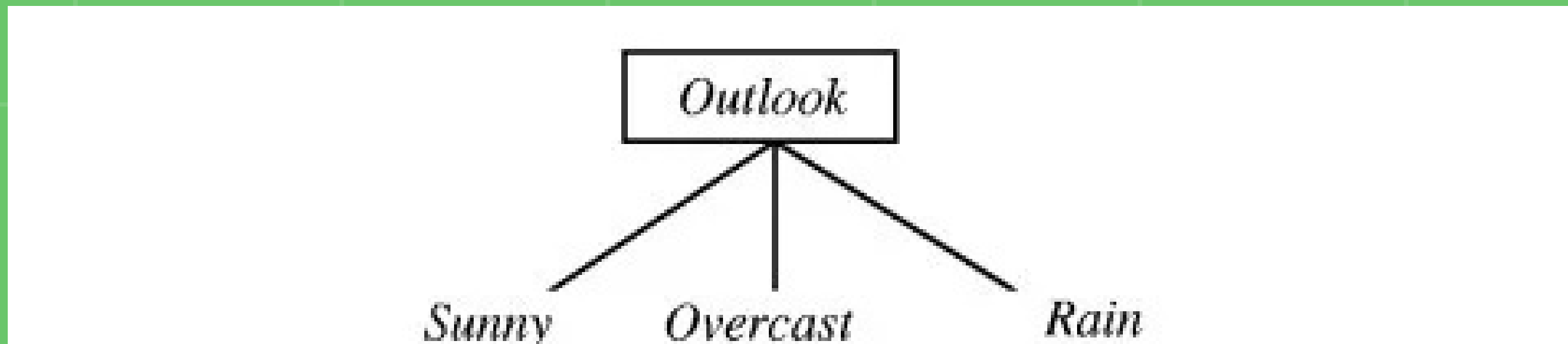
4- Windy:

$$\text{Gain(Windy)} = 0.94 - 0.892 = \mathbf{0.048}$$

Decision Tree Example

First Step: choose the root node

The Root Node is Outlook



Decision Tree Example

Second Step: For Each Subset of the root

Recalculate Entropy

Root subsets (branches): sunny, overcast, rain

First subset (sunny): (5 records → 2 Yes, 3 No)

$$Entropy(Sunny) = - \left(\frac{2}{5} \log_2 \frac{2}{5} + \frac{3}{5} \log_2 \frac{3}{5} \right) = 0.971$$

Decision Tree Example

Second Step: For Each Subset of the root
Recalculate Entropy

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Decision Tree Example

Second Step: For Each Subset of the root, Recalculate Entropy

Feature(1): Temperature

Variables: Hot, Mild, Cool

1- Entropy for Hot (2 samples: 2 no).

Because both are no so entropy = 0

2- Entropy for Mild (1 sample: 1 yes).

Entropy = 0

3- Entropy for Cool (2 samples: 1 yes, 1 no).

Because both are different so entropy = 1

Decision Tree Example

Second Step: For Each Subset of the root, Recalculate Entropy

Feature(1): Temperature

Variables: Hot, Mild, Cool

Weighted entropy for Temperature with subset(sunny):

$$Entropy(Temp) = \frac{2}{5} \cdot 0 + \frac{1}{5} \cdot 0 + \frac{2}{5} \cdot 1 = \mathbf{0.4}$$

Decision Tree Example

Second Step: For Each Subset of the root, Recalculate Entropy

Feature(2): Humidity

Variables: High, Normal

1- Entropy for High (3 samples: 3 no)

$$\text{Entropy} = 0$$

2- Entropy for Normal (2 samples: 2 yes)

$$\text{Entropy} = 0$$

Weighted entropy for Humidity with subset(sunny):

$$\text{Entropy}(\text{Humidity}) = \frac{3}{5} \cdot 0 + \frac{2}{5} \cdot 0 = \mathbf{0.0}$$

Decision Tree Example

Second Step: For Each Subset of the root, Recalculate Entropy

Feature(3): Windy

Variables: False, True

1- Entropy for False (3 samples: 2 no, 1 yes)

$$\text{Entropy} = 0.918$$

2- Entropy for True (2 samples: 2 yes)

$$\text{Entropy} = 0$$

Weighted entropy for Windy with subset(sunny):

$$\text{Entropy}(\text{Windy}) = \frac{3}{5} \cdot 0.918 + \frac{2}{5} \cdot 0 = \mathbf{0.951}$$

Decision Tree Example

Second Step: For Each Subset of the root, Recalculate Entropy

Choosing the feature with the highest information gain

1- Temperature:

$$0.971 - 0.400 = 0.571$$

2- Humidity:

$$0.971 - 0.000 = 0.971$$



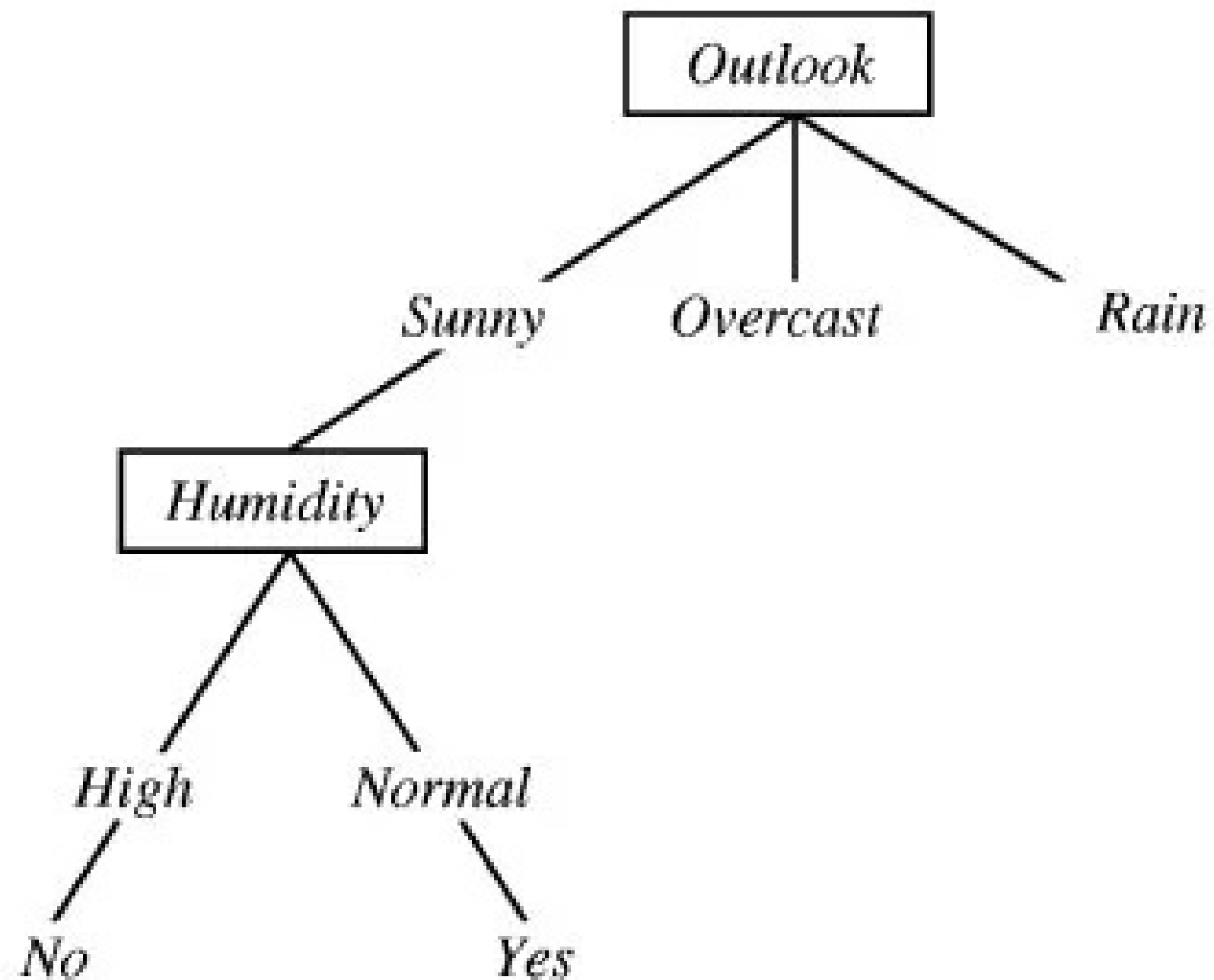
3- Windy:

$$0.971 - 0.951 = 0.020$$

Decision Tree Example

Second Step: For Each Subset of the root, Recalculate Entropy

The Best Feature to Split "Sunny" Subset is **"Humidity"**:



Decision Tree Example

Second Step: For Each Subset of the root

Recalculate Entropy

Root subsets (branches): sunny, overcast, rain

Second subset (rain): (5 records → 3 Yes, 2 No)

$$\textit{Entropy}(\textit{Rainy}) = - \left(\frac{3}{5} \log_2 \frac{3}{5} + \frac{2}{5} \log_2 \frac{2}{5} \right) = \mathbf{0.971}$$

Decision Tree Example

Second Step: For Each Subset of the root
Recalculate Entropy

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Decision Tree Example

Second Step: For Each Subset of the root, Recalculate Entropy

Feature(1): Temperature

Variables: Mild, Cool

1- Entropy for Mild (3 samples: 2 yes, 1 no).

$$\text{Entropy} = 0.918$$

2- Entropy for Cool (2 sample: 1 yes, 1 no).

$$\text{Entropy} = 1$$

Weighted entropy for Temperature with subset(rainny):

$$\text{Entropy}(Temp) = \frac{3}{5} \cdot 0.918 + \frac{2}{5} \cdot 1 = \mathbf{0.951}$$

Decision Tree Example

Second Step: For Each Subset of the root, Recalculate Entropy

Feature(2): Humidity

Variables: Normal, High

1- Entropy for Normal (4 samples: 3 yes, 1 no)

$$\text{Entropy} = 0.811$$

2- Entropy for High (1 sample: 1 no)

$$\text{Entropy} = 0$$

Weighted entropy for Humidity with subset(rainny):

$$\text{Entropy}(\text{Humidity}) = \frac{4}{5} \cdot 0.811 + \frac{1}{5} \cdot 0 = \mathbf{0.951}$$

Decision Tree Example

Second Step: For Each Subset of the root, Recalculate Entropy

Feature(3): Windy

Variables: False, True

1- Entropy for False (3 samples: all yes).

$$\text{Entropy} = 0$$

2- Entropy for True (2 sample: both no).

$$\text{Entropy} = 0$$

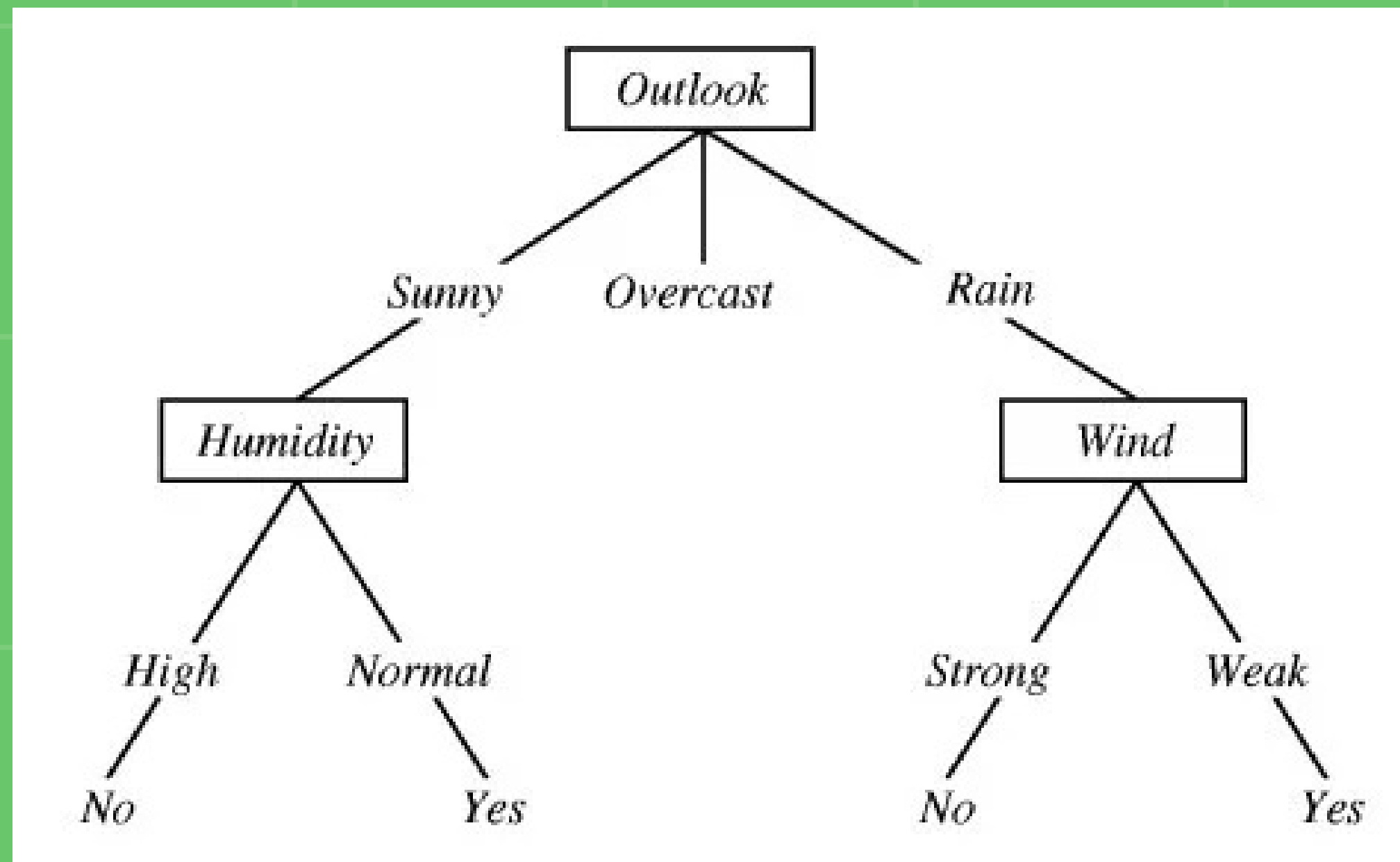
Weighted entropy for Windy with subset(rainny):

$$\text{Entropy}(\text{Windy}) = \frac{3}{5} \cdot 0 + \frac{2}{5} \cdot 0 = \mathbf{0.0}$$

Decision Tree Example

Second Step: For Each Subset of the root, Recalculate Entropy

As Windy has the highest information gain, it will be the next split for rainy.



Decision Tree Example

Second Step: For Each Subset of the root

Recalculate Entropy

Root subsets (branches): sunny, overcast, rain

Last subset (overcast): (4 records → all yes)

As all samples are yes then entropy = 0 , so we will only have one branch with yes!

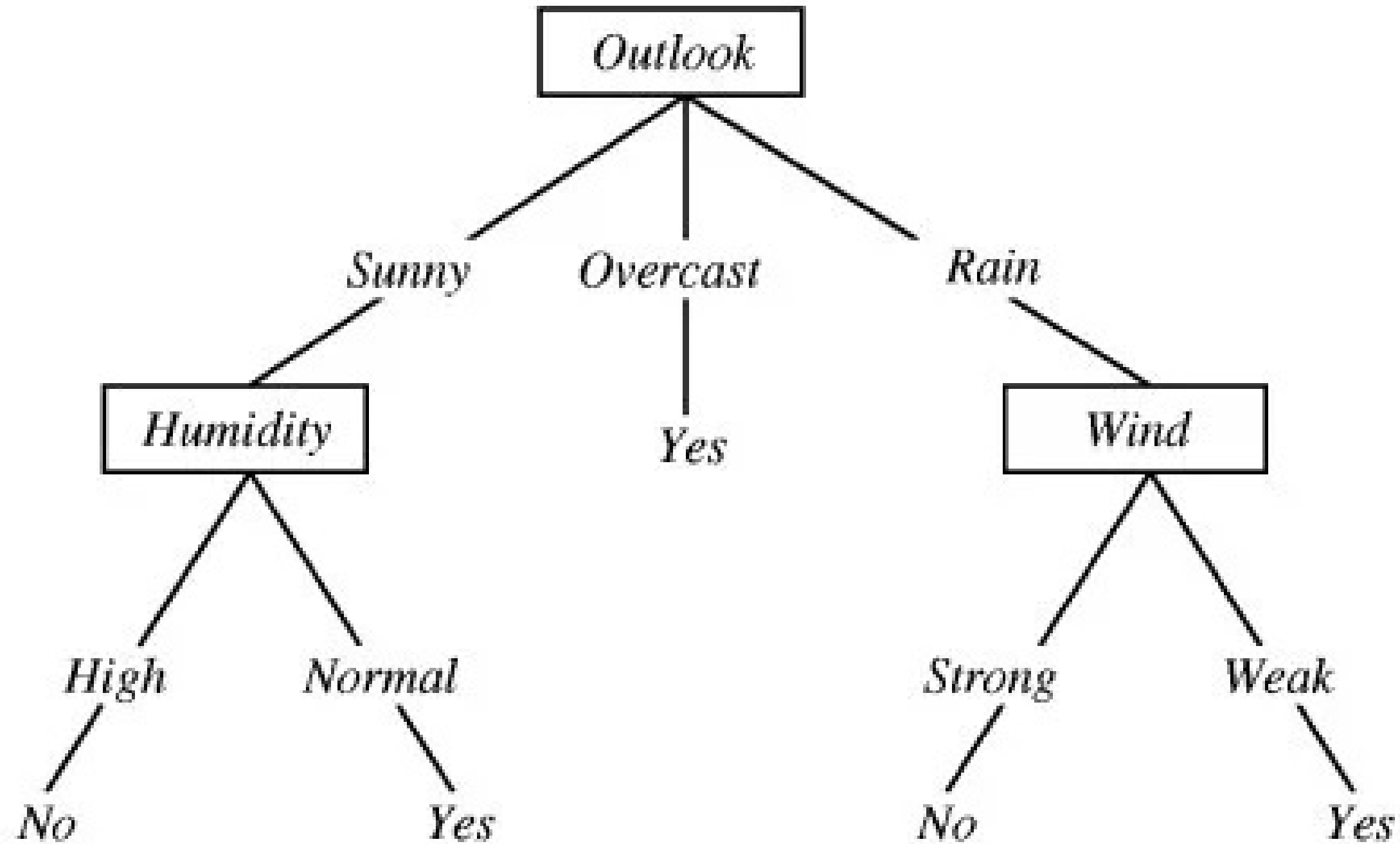
Decision Tree Example

Second Step: For Each Subset of the root
Recalculate Entropy

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Decision Tree Example

Final Tree



When to stop splitting?

- All data in a node belongs to the same class.
- You set a max depth.
- No more features to use

It is your turn

Can you construct a tree with this dataset?

<i>RID</i>	<i>age</i>	<i>income</i>	<i>student</i>	<i>credit_rating</i>	<i>Class: buys_computer</i>
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31 . . . 40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31 . . . 40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31 . . . 40	medium	no	excellent	yes
13	31 . . . 40	high	yes	fair	yes
14	>40	medium	no	excellent	no



How Regression Trees Work?

Regression Tree

- A regression tree follows the same steps as a classification tree, but with a few key differences. Instead of calculating entropy, we calculate the **squared error**, which measures how far the predicted values are from the actual values. And instead of using the Information Gain to choose the best split, we look at the **sum of squared errors (SSE)**. The split that gives us the lowest SSE is chosen as the next split in the tree.

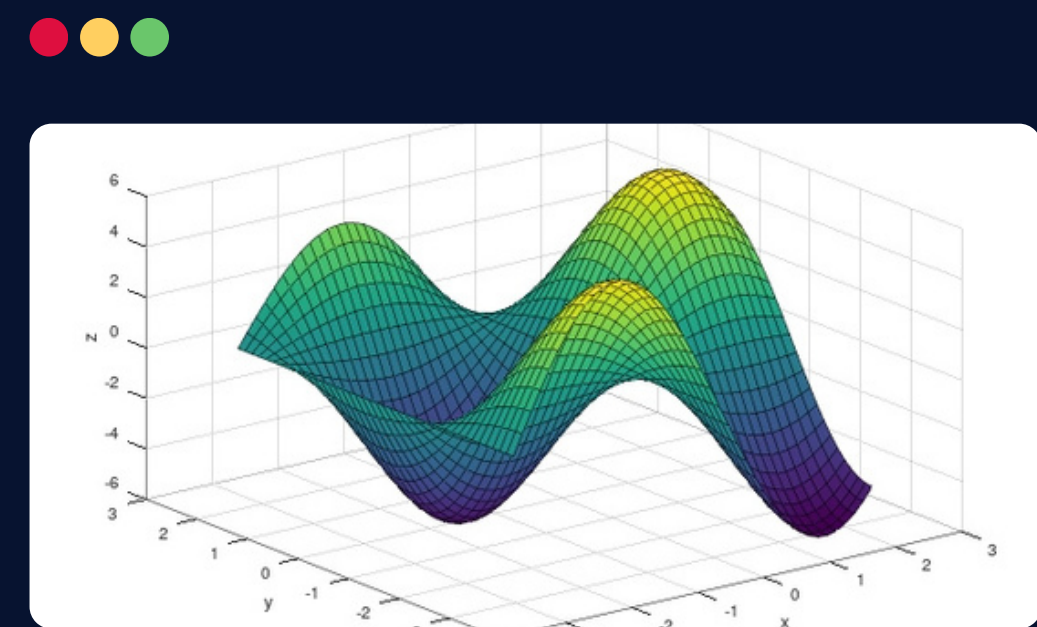
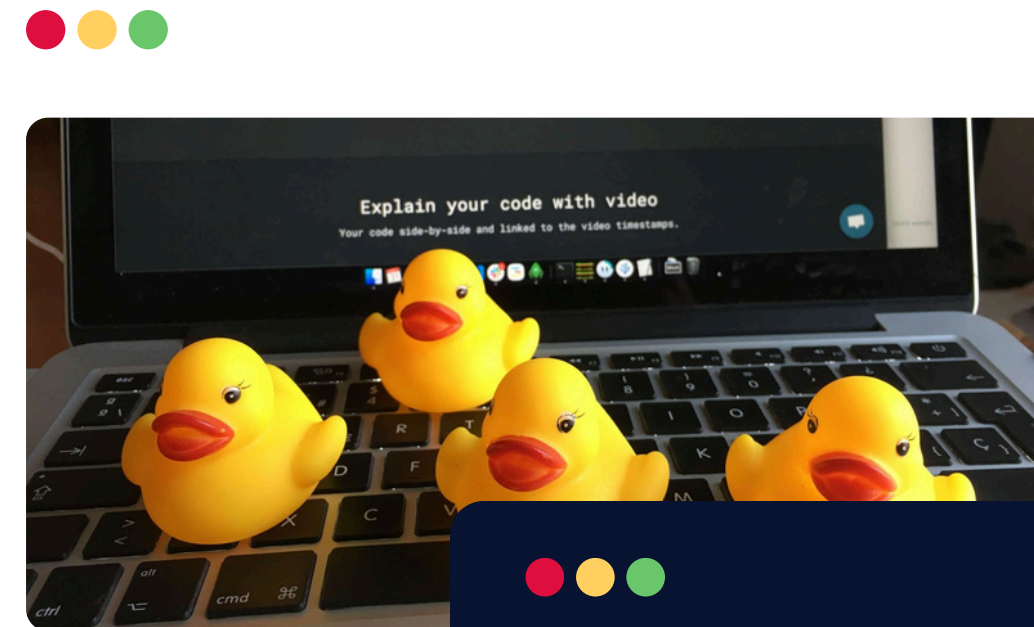
Regression **vs** Classification Tree

Step	Classification	Regression
Impurity Measure	Entropy	Squared Error
Splitting Metric	Information Gain	Sum of Squared Error (SSE)
Split Decision	Maximize Information Gain	Minimize Sum of Squared Error

Example of Regression Tree



[Click Here](#)



Thank You

created by:
Ahd Ehab